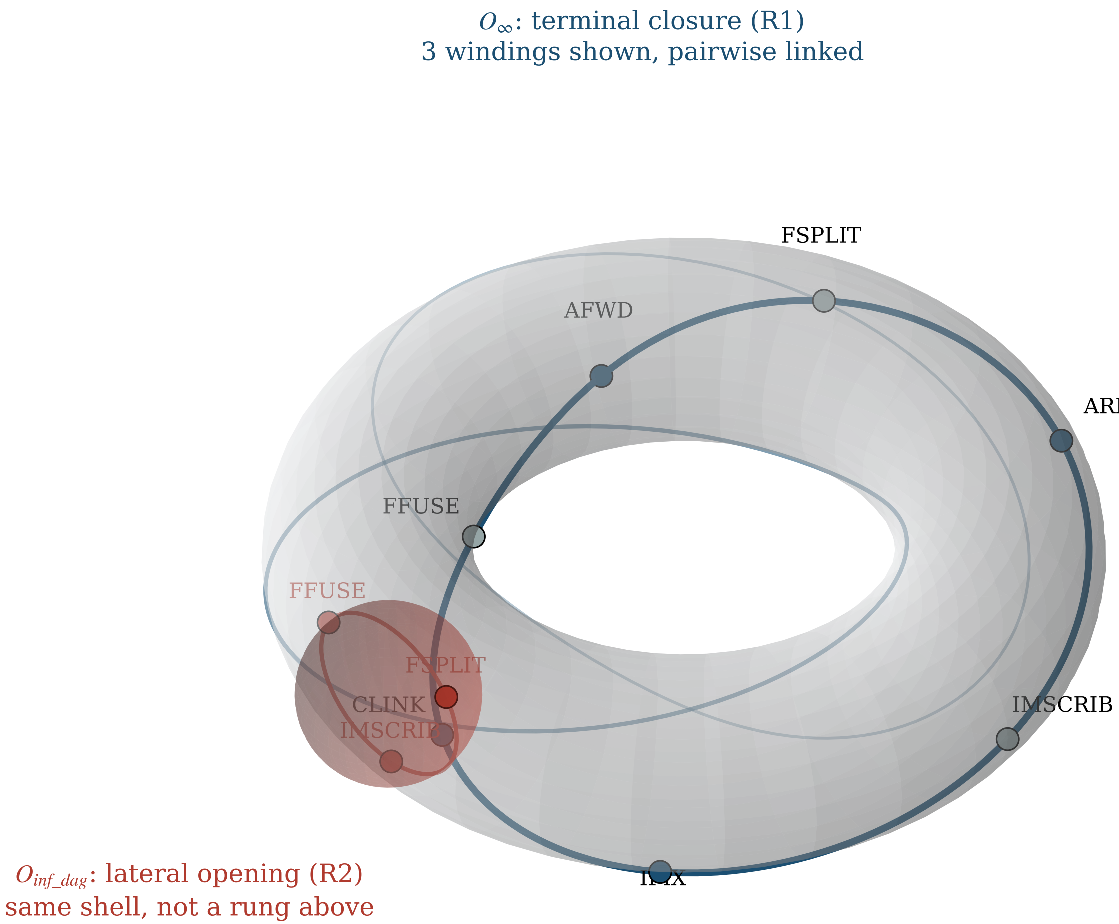
Graph execution as a hopfionic loop bundle
the bootstrap cycle's windings, linked, vs. the lateral O_{inf_dag} opening



Each strand is a $(1,1)$ curve on a shared torus (one loop toroidally and poloidally at once); phase-offset strands are pairwise linked, the same interleaving that links Hopf-fibration circle bundles — the visual reading of re-entering a closed program (winding_count) from a different phase. Grey nodes are no-work opcodes (VINIT/TANCH/IMSCRIB/FSPLIT/FFUSE); blue nodes do real work. The red sphere is the R2 replicative-opening program: its tangent loop at the departure point, rotated 180° about its own diameter, sweeping out a bounded spherical region that genuinely carves into the torus (~22% of the sphere's surface lies inside it) rather than merely touching it at one point.